Supporting Information

Beyond Candidate Generation: A Validation-First Protocol for AI-Assisted Transition-State Discovery

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S1. Protocol checklists

S1.1 P0: chemical preflight

- Reactive pairs: which bonds form and which break, with their actual distances in the starting geometry.
- Charge, multiplicity, and every accessible d^n spin state, including broken-symmetry solutions.
- Connectivity: build a bond graph with a distance criterion; check the intended bonds and any unplanned bond.
- Conformers: pre-fold flexible ligands/chains; retain several low-energy conformers.
- Reference state: reactant or encounter complex and product, at the same level as the TS.
- Candidate filter: hard overlaps, clash ratios, heavy-atom RMSD, connectivity changes.

S1.2 P1: barrier-region scan

- Use 4–8 points along the reaction coordinate, not a blind OptTS.
- Each point is an independent constrained optimization, starting from the previous converged geometry.
- Only converged points enter the profile; the highest converged point is the TS guess.
- Check the profile for monotonicity, shallow double bumps, and flat regions.

S1.3 P2: constrained reference and Hessian

- Run a constrained `Opt + Freq` at the chosen distance.
- The lowest modes must involve the declared reactive pair; otherwise do not run OptTS.
- Keep the Hessian and the reference geometry; both are reused downstream.

S1.4 P3: OptTS

- Read the P2 Hessian (`InHess Read` in ORCA; `readfc` in Gaussian).
- If displacing, scale by the target bond-length change ($\sim$0.05 Å), not the maximum atomic displacement.
- Apply the stop rules in Table 2.

S1.5 P4: validation

- Exactly one significant imaginary frequency.
- The imaginary mode moves the declared reactive pair; localization is not below 15%.
- Key bond lengths match the intended step.
- IRC or downhill relaxation reaches the intended reactant and product.
- $\langle S^2 \rangle$ and Mulliken spin are consistent with the geometry.
- Optimization/frequency/IRC at the same level; ΔE for minimum-vs-saddle; one solvation convention.

S2. Input templates

S2.1 ORCA: per-point constrained optimization

```
! Opt TPSS D3BJ def2-SVP RIJCOSX
%pal nprocs 12 end
%maxcore 1100
%cpcm
  smd true
  SMDsolvent "Toluene"
end
%geom
  MaxIter 220
  Constraints
    { B 20 56 1.750 C }
  end
end
* xyzfile 0 2 start.xyz
```

S2.2 ORCA: constrained reference + frequency

```
! Opt Freq TPSS D3BJ def2-SVP RIJCOSX
%pal nprocs 12 end
%maxcore 1100
%cpcm
  smd true
  SMDsolvent "Toluene"
end
%geom
  MaxIter 220
  Constraints
    { B 20 56 1.750 C }
  end
end
* xyzfile 0 2 start.xyz
```

S2.3 ORCA: OptTS reading the reference Hessian

```
! OptTS Freq TPSS D3BJ def2-SVP RIJCOSX
```

```
%pal nprocs 12 end
%maxcore 1100
%cpcm
  smd true
  SMDsolvent "Toluene"
end
%geom
  MaxIter 160
  InHess Read
  InHessName "L1-constOptFreq.hess"
end
* xyzfile 0 2 L1-constOptFreq.xyz
```

S2.4 ORCA: IRC

```
! IRC TPSS D3BJ def2-SVP RIJCOSX
%pal nprocs 12 end
%maxcore 1100
%irc
  MaxIter 20
  InitHess Read
  Hess_Filename "L2-TSopt.hess"
  Direction both
  PrintLevel 2
  Trajectory True
end
* xyzfile 0 2 L2-TSopt.xyz
```

S2.5 Gaussian: constrained reference + frequency

```
%mem=48GB
%nprocshared=24
%chk=01_const_optfreq.chk
#p UB3LYP/6-31G(d) opt=(calcfc,modredundant,maxcycles=220) freq
   em=gd3bj scrf=(smd,solvent=toluene) int=ultrafine

TS step 1: constrained Opt+Freq

0 2
<coordinates>

B 1 2 1.500 F
B 3 4 2.100 F
```

S2.6 Gaussian: OptTS reading the reference checkpoint

```
%mem=48GB
%nprocshared=24
%oldchk=01_const_optfreq.chk
%chk=02_ts_optfreq.chk
#p UB3LYP/6-31G(d) opt=(ts,readfc,noeigen,maxcycles=220) freq
   em=gd3bj scrf=(smd,solvent=toluene) int=ultrafine
```

```
   geom=check guess=read

TS step 2: OptTS+Freq

0 2
```

S2.7 Slurm template

```
#!/bin/bash
#SBATCH -J ts
#SBATCH -N 1
#SBATCH -n 24
#SBATCH -p <partition>
#SBATCH -o slurm-%j.out
#SBATCH --time=48:00:00

export g16root='<g16root>'
source "${g16root}/g16/bsd/g16.profile"
export GAUSS_SCRDIR=/tmp

date
bash run_chain.sh
echo "EXIT=$?"
date
```

S3. Case-study details

The systems in the main-text case studies are 84–98 atom Ni/Fe complexes with organic ligands, treated with TPSS-D3BJ/def2-SVP (RIJCOSX, SMD) for geometry and frequency calculations, and higher-level single points where noted. The numbers below are the observations discussed in the main text.

Table 1: Internal case-study observations.

Case	Observation	Lesson
Per-point scan	16/16 points converged after three collective scans failed; the problematic point converged in 44 cycles	A local unit of failure is more robust than a collective scan
Mode displacement	0.18 Å maximum displacement changed the reactive bond by −0.31 Å; 183 cycles into the reactant basin	Scale by the target bond change (∼0.05 Å)
Second-order saddle	Two rounds returned the same geometry and energy (0.003 kcal mol^{-1}); a finer grid confirmed the soft mode	Stop after the second repeat; disclose or rigidify
Conformer-dominated R/S	Ligand-overlay RMSD 1.62 Å between R and S intermediates	Search conformers before claiming stereoselectivity
Generative model OOD	Four candidates passed the filter but had 3.6–4.4 Å RMSD to the reference	Test the applicability domain on the system class

S4. AI candidate generation: practical notes

S4.1 UniTS-Gen

UniTS uses a 0-based `reactive_atom_idx`, whereas many post-processing scripts use 1-based atom indices. Convert explicitly. The intended use is multi-sampling: generate ten candidates, prefilter, and send only a few to DFT. Single-shot correctness is moderate; the published benchmark reports that multi-sampling recovers all five previously failed cases. In an internal test on an 84-atom Ni broken-symmetry system, four candidates passed the geometric filter but had 3.6–4.4 Å heavy-atom RMSD to the reference; the model was out of distribution for that system class.

S4.2 ORCA GOAT-TS

GOAT-TS is a built-in, non-ML search. A minimal input is:

```
! GOAT-TS B3LYP D3BJ def2-SVP TightSCF
%pal nprocs 12 end
%maxcore 1100
%GOAT
  USEGOAT true
  MAXITER 30
  REACTIVEATOMS {0 4 5} end
end
* xyz 0 1
<coordinates>
*
```

`REACTIVEATOMS` uses braces and 0-based indices. GOAT is a search tool; its output still goes through P2–P4. It becomes expensive for large flexible systems and is best used for channel discovery or medium-sized systems.

S5. Stop rules

Table 2: Stop rules used in the protocol.

#	Symptom	Action
1	Lowest mode at P2 does not involve the reactive pair	Do not run OptTS; change constraint/conformer/guess
2	OptTS energy decreases monotonically for > 50 cycles while the reactive bond is unchanged	Wrong basin; stop and return to P1/P2
3	Two displaced OptTS runs return to the same geometry and energy (< 0.01 kcal mol^{-1})	Stop; accept with disclosure, rigidify, or run IRC along the reactive mode
4	Two or three shallow imaginary modes around -15 to -25 cm^{-1}	Flat transition valley; use the last geometry's frequency as a reference barrier and say so
5	Imaginary-mode localization below 15%	Delocalized soft mode; not the target coordinate
6	`SADDLE-1` and the declared pair matches, but key bond lengths are wrong	Wrong channel; do not report this barrier
7	IRC or relaxation endpoint is not the intended reactant/product	No barrier claim; return to P1/P2
8	Single-point $\langle S^2 \rangle$ differs from the geometry job	Wrong electronic state; repeat the full spin setup
9	R/S difference without a conformer search	Not a stereoselectivity result; do the ensemble first
10	xtb and DFT disagree	DFT wins; xtb is for folding only